

Module 2

Foundations of Computing

Computer Science Department · School of ICT · Copperbelt University

LECTURE OVERVIEW — Module 2

Three sections — building from data concepts to computing history to device classification:

A

1.1 Data, Information, Knowledge & Wisdom

DIKW Hierarchy · Qualitative vs. Quantitative · Structured vs. Unstructured · Data Processing Cycle · CART Quality Framework

B

1.2 History and Evolution of Computing

5 Generations of Computers · Moore's Law · Mainframes → Microcomputers → Smartphones → Wearables

C

1.3 Modern Classification of Computers

Supercomputers · Mainframes · Servers · Workstations · PCs · Mobile · Embedded Systems · IoT

MODULE 2 · SECTION A

Section 1.1

Data, Information, Knowledge & Wisdom

DIKW Hierarchy · Data Types · Processing Cycle · Information Quality

What is Data? — The Raw Material of Computing

Data: Raw, unprocessed facts, symbols, numbers, or observations — without context or interpretation. Data is the most fundamental level of the information hierarchy. By itself, data has NO meaning.

123 Numeric

36.5 · 2048 · 0.001

ABC Text / String

"Kitwe" · "Zambia" · "CS120"

01 Binary

01000001 · 11001100 · 10110101



Date / Time

2025-03-24 · 14:22:07



Image / Pixel

RGB(255, 87, 51) · 640×480 px

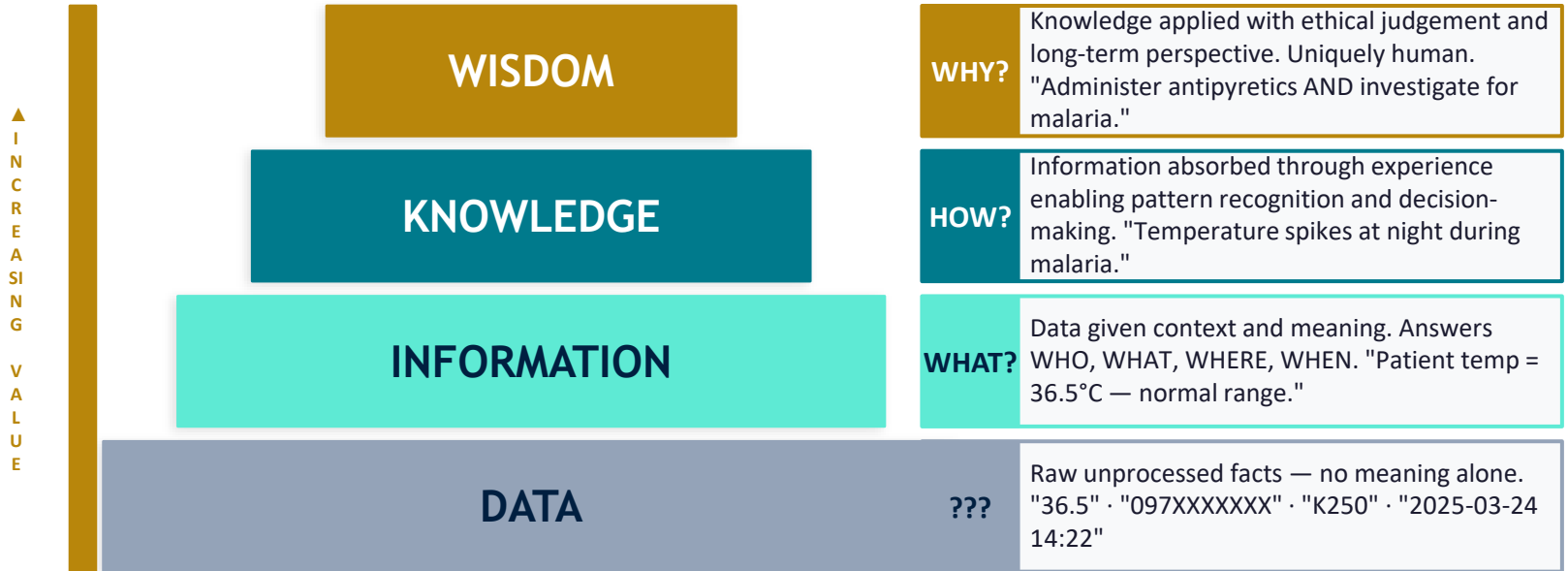


Audio Sample

44100 Hz · 16-bit · amplitude

KEY INSIGHT: Every piece of data stored in a computer — no matter how complex — ultimately reduces to binary digits (bits): 0s and 1s. This is why understanding number systems (Module 3) is the next step.

The DIKW Hierarchy — From Raw Facts to Wisdom



Zambia Example: "K250 · 097xxx" → DATA "K250 sent to John 2:22pm" → INFO "John pays every 25th" → KNOW "Plan cash flow by salary date" → WISDOM

5 / 33

GIGO — Garbage In, Garbage Out: A system produces output no better than its input. If your data is wrong, all downstream information, knowledge, and decisions are wrong too.

Qualitative vs. Quantitative Data

QUALITATIVE DATA		QUANTITATIVE DATA
Definition	Descriptive, non-numeric. Captures qualities, opinions, and characteristics.	Numeric data that can be measured, counted, and used in mathematical operations.
Answers	Why? How? What kind? What does it feel like?	How many? How much? How often? What is the rate?
Types	Nominal: categories with no order (blood types, gender, ethnicity) Ordinal: ordered categories (exam grades: A, B, C, D, F)	Discrete: countable whole numbers (50 students, 3 errors, 7 modules) Continuous: any value in a range (temperature, weight, voltage)
Collection	Interviews, focus groups, observation, open-ended surveys, document analysis	Experiments, sensors, measurements, counting, closed surveys, financial records
Analysis	Thematic coding, content analysis, narrative analysis	Statistics: mean, median, mode, standard deviation, regression, ML models
CBU Example	Student feedback on lecture quality: 'The lecturer explains clearly and uses relevant examples'	Student performance data: 'Average mark: 62.4% · Highest: 94% · Fail rate: 11%'

Both types are equally valid — different questions require different data. Good research design specifies WHICH type is needed before collection begins.

Structured, Semi-Structured & Unstructured Data

Of all data produced globally (IDC, 2023):

20%
Structured

10%
Semi

70% Unstructured

Structured

Organised in predefined rows and columns. Every record has the same fields. Easy to search and analyse with SQL.

Storage:

Relational databases: MySQL, PostgreSQL, Microsoft SQL Server, Oracle, Microsoft Access

Examples:

- Student record: (ID, Name, Programme, Year, GPA)
- Bank transaction: (TxnID, Amount, Date, AccountNo)
- Zambia census data: (Province, District, Population, Age)



Semi-Structured

Has some organising elements (tags, keys) but not the rigid schema of relational databases. Flexible structure.

Storage:

Document stores: MongoDB, DynamoDB, Elasticsearch, CouchDB. Also: JSON files, XML files, CSV exports

Examples:

- JSON from MTN MoMo API: {txn_id, amount, sender, timestamp}
- HTML web page: has structure but not rigid schema
- Email with metadata: (From, To, Subject) but free-form body



Unstructured

No predefined format or schema. Requires AI/ML tools to extract meaning. The majority of all data produced.

Storage:

Data lakes, object storage: Amazon S3, Azure Data Lake, Google Cloud Storage, Hadoop HDFS

Examples:

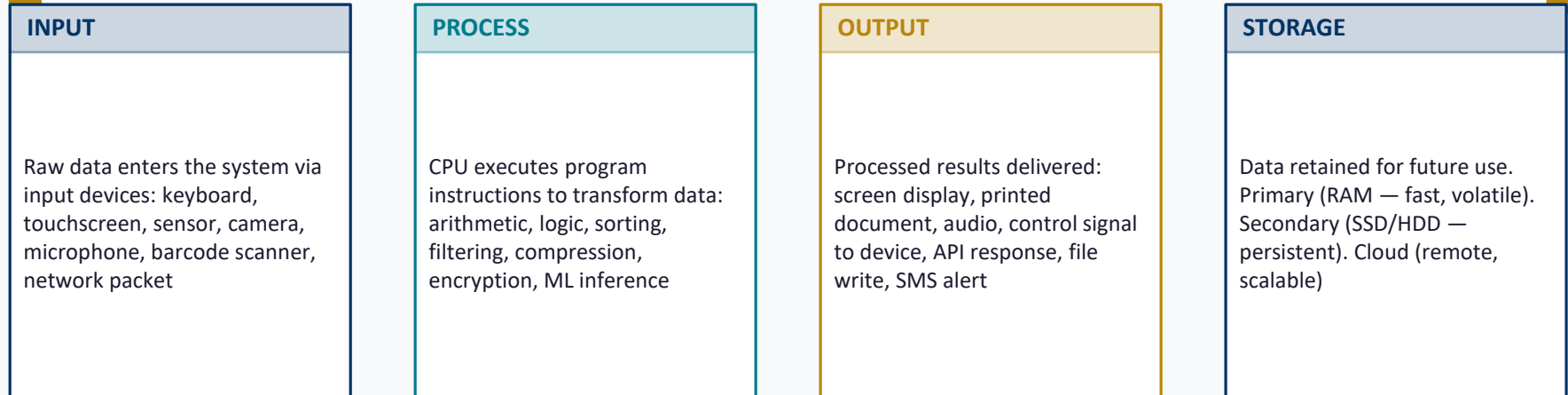
- WhatsApp messages and voice notes
- Satellite images of Copperbelt copper mines
- Security camera footage from Lusaka City Market
- Medical imaging: X-rays, CT scans, MRI files

The Data Processing Cycle

Every computing system — from a pocket calculator to a cloud AI platform — follows this universal model.



◀ **FEEDBACK** — Outputs and stored results continuously refine future inputs, processing rules, and system design



Data Processing Cycle – Zambian Real-World Examples

Apply the $I \rightarrow P \rightarrow O \rightarrow S \rightarrow F$ model to these systems you encounter in Zambia every day:

System	Input	Process	Output	Storage / Feedback
MTN MoMo Transfer	Sender phone number, recipient number, amount, PIN confirmation	Authenticate PIN; verify sender balance; check recipient number; compute fees	Transaction confirmation SMS; debit/credit applied; receipt	Transaction log in banking database → fraud model retrained on transaction patterns
Zambia Police Fingerprint ID	Fingerprint scan from biometric reader	Match against national fingerprint database using ridge pattern algorithm	Match found / not found; suspect identity confirmed or cleared	New fingerprint added to national DB → system accuracy improves with each new record
Smart Farming Irrigation (IoT)	Soil moisture sensor reading every 30 minutes	Compare moisture level to optimal crop range threshold set by agronomist	Pump activation signal if moisture < threshold; alert SMS to farmer	Historical moisture/yield data stored → optimal thresholds refined each season
LMS Online Exam	Student login credentials; answers to MCQ and essay questions; system timestamp	Authenticate student; compare answers to answer key; apply marking rubric; timer check	Immediate score display; grade submitted to academic registry; feedback to student	All answers stored in LMS DB → analytics dashboard identifies weak question areas for lecturers
Zambia E-Passport Application	Applicant personal data, biometrics (photo, fingerprints), supporting documents	Cross-check NRC number; validate documents; schedule appointment; compute fees	Application confirmation; appointment booking; payment receipt; passport issued	Passport database updated → pattern analytics detect fraudulent applications

Measuring Information Quality – The CART Framework

GIGO in reverse: Good information requires that each of four quality dimensions is satisfied. Failure in ANY ONE undermines decisions built on that information.

A

ACCURACY

Data correctly represents the real-world entity or event it describes. Inaccuracy can arise from: data entry errors, sensor calibration faults, fraud, transmission errors, or stale data that has not been updated.

- ✓ Patient blood type recorded as 'O-' — confirmed by two independent blood tests
- ✗ Typo: 'K2,500' payment entered as 'K25,000'
- ⚠ Financial loss; overpayment; incorrect medical treatment

C

COMPLETENESS

All required data fields are present and populated. Missing fields can make information unusable or, worse, misleading — decisions made on partial data may be systematically biased.

- ✓ Student emergency contact record: name + phone + relationship + address all filled
- ✗ Student emergency record with no phone number
- ⚠ Cannot reach next-of-kin in an emergency; delayed crisis response

R

RELEVANCE

Information is appropriate and pertinent to the specific purpose or decision at hand. Irrelevant data adds noise, increases cognitive load, and can lead to incorrect conclusions when out-of-context data is misapplied.

- ✓ This quarter's sales data used to evaluate this quarter's performance
- ✗ 2010 census data used to plan 2025 road infrastructure
- ⚠ Wasted resources; infrastructure built in wrong locations

T

TIMELINESS

Information is available when it is needed and reflects the current state of affairs. Some decisions require real-time data; others tolerate daily or monthly reporting. Timeliness requirements vary by use case.

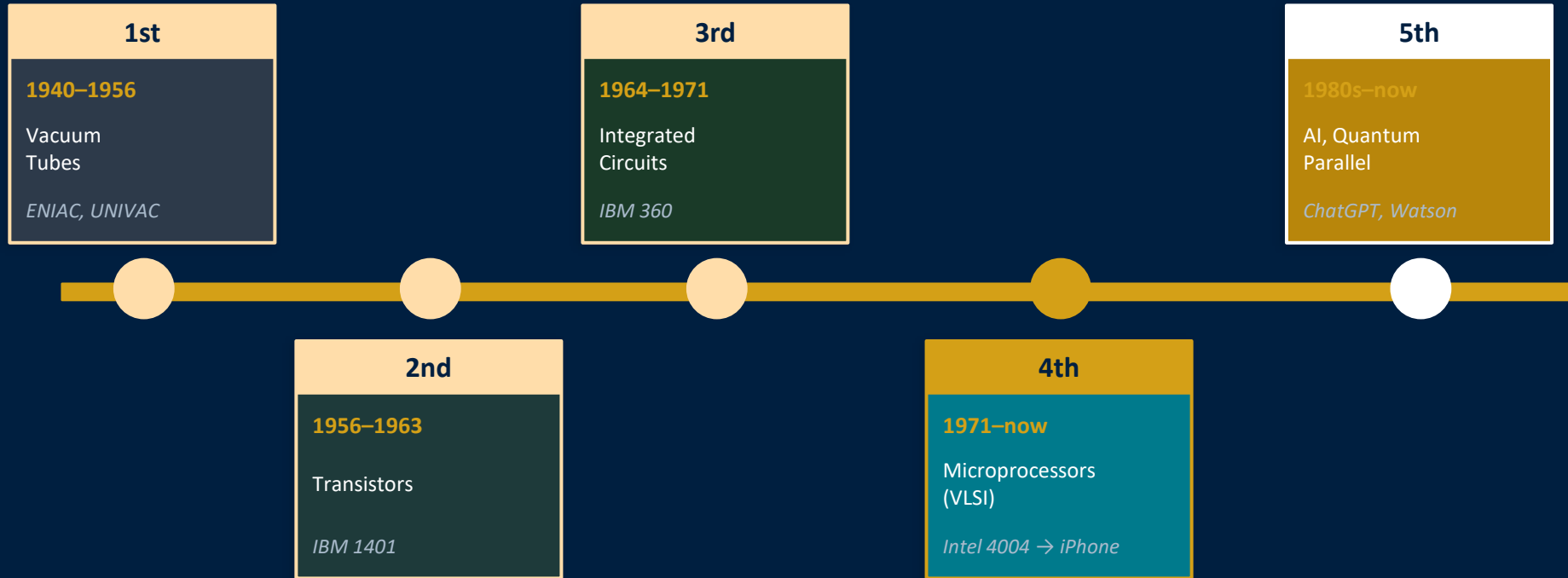
- ✓ Real-time copper price feed for a commodity trading platform
- ✗ Yesterday's MTN network outage report used for today's maintenance schedule
- ⚠ Maintenance deployed where network is already restored; outages missed

Section 1.2

History & Evolution of Computing

5 Generations · Moore's Law · Mainframes to Wearables

5 Generations of Computers – Timeline Overview



Generations 4 & 5 are most relevant to modern computing — detailed on the following slides.

Generations 1-3: Vacuum Tubes → Transistors → ICs

1

1940–
1956

1st Generation — Vacuum Tubes

- ▶ Vacuum tubes: glass cylinders controlling electron flow — each the size of a light bulb
- ▶ ENIAC (1945): 18,000 tubes, 30 tonnes, 1,800 sq ft, consumed 150kW electricity
- ▶ Failures every ~2 hours — teams of engineers maintained them 24/7
- ▶ Programmed in machine code (binary); no operating system; one program at a time
- ▶ Applications: ballistic trajectory calculation (WWII), census data processing

Impact:

*Proved electronic computing was viable.
Created the foundations of computer science as a discipline.*

2

1956–
1963

2nd Generation — Transistors

- ▶ Transistor vs. vacuum tube: smaller, faster, more reliable, ~1/100th the power, far less heat
- ▶ First truly commercial computers for businesses — IBM dominated this era
- ▶ Assembly language introduced; high-level languages emerged: FORTRAN (1957), COBOL (1959)
- ▶ SABRE (1960): first real-time airline reservation system — still architecturally influential today
- ▶ Cost fell from millions to hundreds of thousands of USD per machine

Impact:

*Created the commercial IT industry.
Banks, airlines, and government agencies adopted computing for first time.*

3

1964–
1971

3rd Generation — Integrated Circuits (ICs)

- ▶ IC = multiple transistors etched onto a single silicon chip. First: 4 transistors; by 1971: thousands
- ▶ Operating systems introduced: IBM OS/360 — one computer could run many programs concurrently
- ▶ Input/output revolution: keyboards and monitors replaced punch cards
- ▶ IBM System/360 (1964): first 'family' of compatible computers — a single architecture, multiple price points

Impact:

Computers reached universities and research institutions. The software industry began to emerge as separate from hardware.

4th Generation (1971-Present): The Microprocessor Era

The 4th generation is defined by VLSI (Very Large Scale Integration) — placing millions, then billions, of transistors on a single chip.

1971 Intel 4004

First commercial microprocessor (2,300 transistors, 10 μm process). Complete CPU on one chip. Changed everything.

1975 Altair 8800

First personal computer kit. Inspired Bill Gates and Paul Allen to found Microsoft.

1981 IBM PC

Mass-market personal computer. MS-DOS operating system. Defined the modern PC standard. 16-bit architecture.

1984 Apple Macintosh

First mainstream GUI with mouse, icons, and windows. Made computers accessible to non-technical users.

1991 World Wide Web

Tim Berners-Lee released HTTP and HTML. The internet became the web. Information access transformed globally.

2007 Apple iPhone

ARM SoC, capacitive touchscreen, app ecosystem, GPS. Redefined what a computer is. Zambia's mobile-first era begins.

2016 Multi-core + GPU AI

Deep learning explosion. GPUs repurposed for neural networks. CPUs with 16+ cores become standard. Cloud dominates.

2020+ Apple M1 / NPU Era

3–5 nm chips. Neural Processing Units built into SoCs for on-device AI. 5G. 28+ billion transistors per chip.

5th Generation (1980s-Present): AI, Parallel Processing & Quantum

Unlike previous generations, the 5th is defined not by a single hardware technology but by the APPLICATION of intelligence. We are currently in the EARLY stages of this era.

Artificial Intelligence & Machine Learning

- ▶ Systems that LEARN from data rather than following explicit programmed rules
- ▶ Machine learning: neural networks trained on massive datasets (images, text, speech)
- ▶ Examples: IBM Watson (medical diagnosis), Google AlphaFold (protein structure prediction), ChatGPT (language generation), fraud detection in banking
- ▶ All large Zambian banks and MTN now use ML for fraud detection on mobile money transactions

Parallel Processing & HPC

- ▶ Distributing computation across thousands of processors simultaneously
- ▶ GPU computing: graphics cards repurposed for AI training (NVIDIA A100: 6,912 CUDA cores)
- ▶ Applications: weather forecasting (requires petaflops), drug discovery, climate modelling, training GPT-4 required 25,000 NVIDIA A100s for 90 days
- ▶ Zambia link: SADC regional weather forecasts are computed on HPC clusters in South Africa

Natural Language Processing (NLP)

- ▶ Computers that understand, generate, and translate human language
- ▶ Applications: voice assistants (Siri, Google Assistant), machine translation (Google Translate), chatbots, sentiment analysis
- ▶ Limitation: most NLP systems perform poorly on African languages including Bemba, Nyanja, and Tonga — a research opportunity for Zambian CS graduates
- ▶ WhatsApp's end-to-end encryption uses NLP-related cryptographic techniques

Quantum Computing

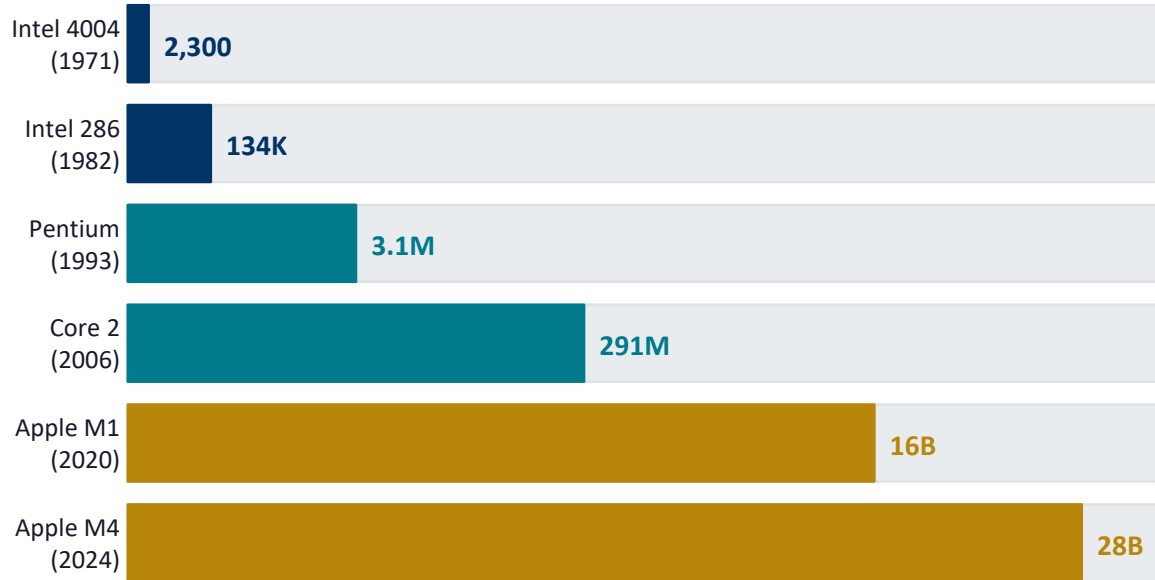
- ▶ Uses quantum bits (qubits) that can exist in superposition of 0 AND 1 simultaneously — enabling exponentially faster computation for specific problem types
- ▶ Current stage: NISQ era (Noisy Intermediate-Scale Quantum) — IBM 433-qubit Osprey (2022), Google Sycamore
- ▶ Potential: optimise logistics networks; simulate molecular interactions for drug design
- ▶ Current reality: still fragile, expensive, and limited to specific algorithm types. Practical general quantum computing remains years away

Moore's Law — The Engine of the Digital Revolution

"The number of transistors on a microchip doubles approximately every two years, while the cost per transistor halves."

— Gordon Moore, Intel Co-Founder (1965, revised 1975)

Transistor Count Growth (Note: this is NOT log-scaled — actual ratios are far more extreme)



Practical Implications

Cost:

1990 PC (\$2,000) < K3,000 smartphone today in power

Size:

Room → desk → pocket → wearable → implant

Performance:

Smartphone > 1990s \$50M supercomputer

New industries:

Cloud, AI, streaming — all depend on Moore

Democratisation:

Cheap smartphones → Zambia's mobile-first era

Slowing?:

<5nm: quantum tunnelling. 3D chips + quantum needed

CALCULATION: Intel 4004 (1971) = 2,300 transistors. Apple M4 (2024) = 28 billion transistors. How many doublings? $28B \div 2,300 = 12.2M \times \rightarrow 2^n = 12.2M \rightarrow n \approx 23.5$ doublings $\div 2$ years = ~ 47 years. ✓ Consistent with 2024–1971=53 years (allowing for slower early years).

From Mainframes to Wearables

1960s–70s Mainframe Era

*Government &
large
corporations
only*

- ▶ IBM System/360 defined enterprise computing — shared by hundreds via dumb terminals
- ▶ Batch processing: jobs submitted in advance; results returned hours later
- ▶ Cost: millions of USD; rooms of equipment; dedicated staff to operate
- ▶ Zambia: early government data processing in Lusaka — IBM mainframes for census and payroll

1980s–90s Personal Computer Era

*Universities,
businesses,
educated
households*

- ▶ IBM PC (1981) + Apple Macintosh (1984): computing moves to the desk
- ▶ Microsoft Windows GUI makes computing accessible to non-technical users
- ▶ Internet: email (early 90s) → World Wide Web (1991) → first websites
- ▶ Zambia: CBU computer labs established mid-1990s; first internet in Zambia 1994

2000s–10s Smartphone Era

*Mass adoption
— >80%
Zambia access
via mobile*

- ▶ Nokia/Motorola feature phones → Apple iPhone 2007 → Android ecosystem 2008
- ▶ Computing in the pocket: camera, GPS, apps, web browser, mobile banking
- ▶ 4G LTE (2012 onwards): video streaming, cloud applications, mobile money
- ▶ Zambia: MTN MoMo (2012) + Airtel Money transforms financial inclusion — mobile-first nation

2020s Wearable & Ambient Era

*Computing
dissolves into
everyday
objects*

- ▶ Smartwatches: Apple Watch, Samsung Galaxy Watch — health monitoring, calls, notifications
- ▶ Smart glasses, AR/VR headsets (Meta Quest, Apple Vision Pro)
- ▶ Implantable: Neuralink brain-computer interface (human trials ongoing)
- ▶ Zambia: smart electricity meters (Zesco), IoT farm sensors, smart mine monitoring

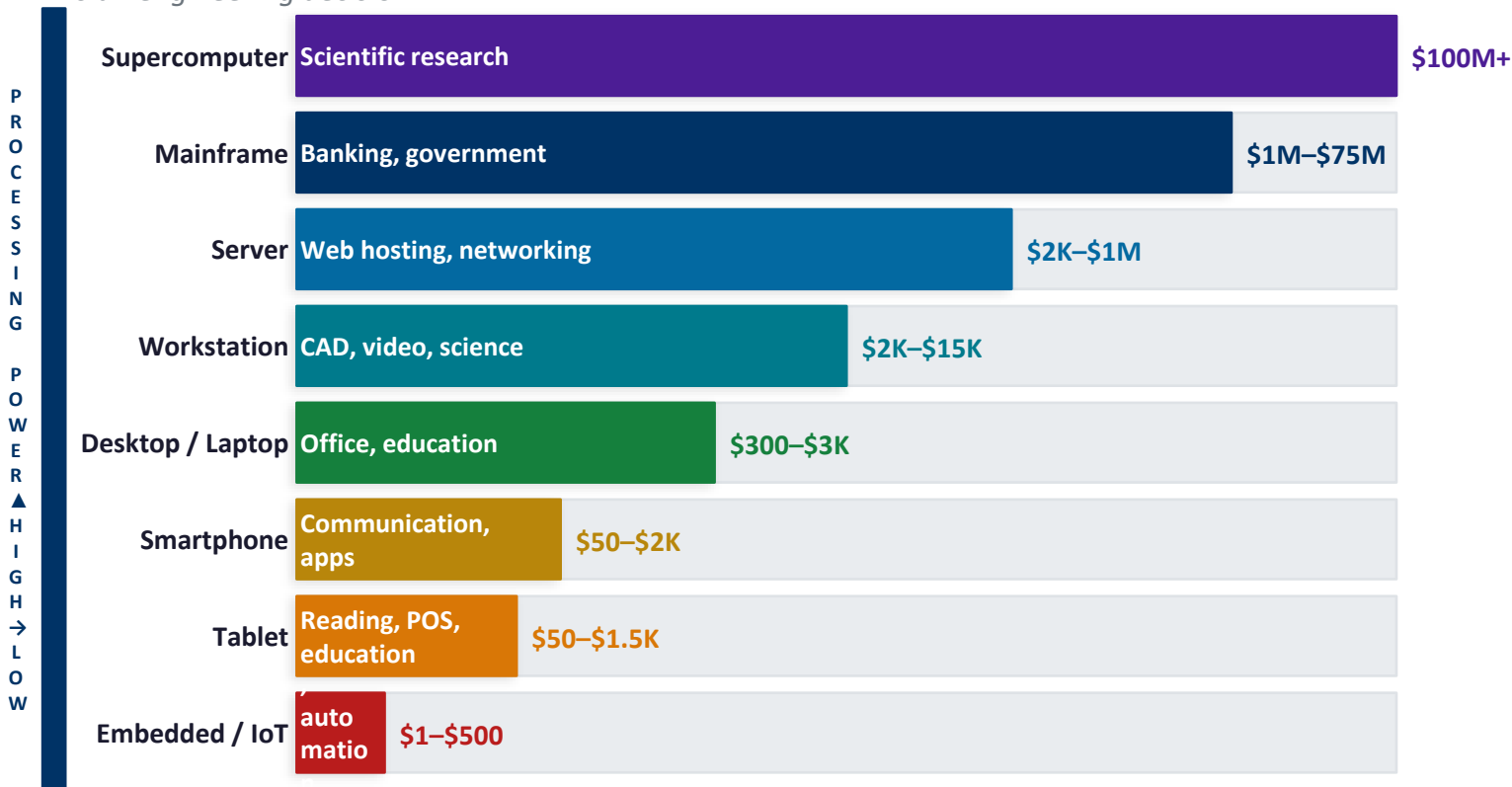
Section 1.3

Modern Classification of Computers

Supercomputers · Mainframes · Servers · PCs · Mobile · IoT · Cloud

Computer Classification — The Spectrum of Computing Power

Computers exist on a spectrum from planetary-scale supercomputers to microscopic embedded chips. Choosing the RIGHT type is an engineering decision.



Supercomputers & Mainframe Computers

Supercomputers

Performance:	Measured in FLOPS. Top machines: Exascale (10^{18} operations/sec). Frontier (USA, 2022): 1.1 ExaFLOPS — first exascale computer.
Architecture:	Tens of thousands of interconnected compute nodes, each with multiple CPUs and GPUs, linked by ultra-high-speed proprietary fabric.
Power:	20–40 megawatts of electrical power — equivalent to a small Zambian town. Cooling is a major engineering and cost challenge.
Cost:	\$100M–\$600M to build; \$10–30M per year to operate. Only governments and major research institutions can afford them.
Applications:	Climate modelling, nuclear simulation, drug discovery (AlphaFold protein folding), earthquake prediction, training large AI models.
Africa:	Cheetah (CSIR, Pretoria, South Africa) is Africa's fastest. Zambia: none currently. Access via cloud HPC services from South Africa.
Ranking:	Top500.org ranks the world's 500 fastest supercomputers — updated biannually using the Linpack benchmark.

Mainframe Computers

Definition:	High-reliability computers for large-scale, high-volume transaction processing — NOT defined by speed, but by reliability and throughput.
Reliability:	'Five nines': 99.999% uptime = ~5 minutes downtime per year. Achieved through redundant components — dual power supplies, cooling, I/O channels.
Throughput:	Process millions of transactions per second. IBM Z16 can process 300 billion transactions per day with on-chip AI fraud detection.
Users:	Supports tens of thousands of concurrent users. Banks, airlines, social security systems — all running simultaneously on shared infrastructure.
Security:	Hardware-enforced security domains prevent cross-application data access. Bank A's data cannot be accessed by Bank B's application on the same machine.
Longevity:	44 of the top 50 global banks and 70% of Fortune 500 companies rely on IBM Z-series mainframes.
Zambia:	Major Zambian banks process retail transactions through mainframe-class infrastructure in regional data centres (South Africa, Zimbabwe, or Lusaka).

Servers — The Backbone of Networked Computing

A server is a computer that provides services, resources, or data to other computers (clients) over a network. Servers run 24/7, are typically headless (no screen/keyboard), and are managed remotely.

Web Server

Serves HTML, CSS, JavaScript, and media to web browsers via HTTP/HTTPS. Apache and Nginx are the most widely deployed.

 CBU's public website runs on a web server. ZICTA's licensing portal. All e-government services.

Application Server

Hosts and executes application logic and business rules. Often sits between the web server and database in a 3-tier architecture.

 MTN MoMo's transaction logic — validates transfers, computes fees, checks fraud rules before communicating with the database.

Database Server

Hosts the DBMS (MySQL, PostgreSQL, SQL Server, Oracle). Processes SQL queries from application servers.

 CBU's student records database. Zanaco's account management system. ZRA's taxpayer database.

Mail Server

Handles email sending (SMTP), receiving (IMAP/POP3), and storage. Your @cbu.ac.zm email lives on a mail server.

 CBU student and staff email. Government .gov.zm domain email infrastructure.

DNS Server

Translates domain names (cbu.ac.zm) to IP addresses — the internet's 'phone book'. Critical infrastructure for all internet access.

 Zambia's national .zm domain is managed by ZICTA. Without DNS, no Zambian website would be accessible by name.

File Server

Centralised file storage accessible to all authorised network users. Enables collaborative working and backup.

 CBU shared network drives for staff. Company document management systems in Zambian corporates.

Workstations & Personal Computers

Workstations

Definition:

High-performance single-user computers for demanding professional applications — occupying the gap between standard PCs and servers.

CPU:

Server-class processors (AMD Threadripper Pro, Intel Xeon) — high core counts (up to 96) optimised for sustained multi-threaded workloads.

RAM:

ECC (Error-Correcting Code) RAM: detects and corrects single-bit memory errors in real-time. Typically 64–512GB. Critical for scientific computation.

GPU:

Professional GPUs (NVIDIA RTX/Quadro series, AMD Radeon Pro) — certified for CAD software, 3D rendering, and ML training workloads.

Applications:

Architectural CAD (AutoCAD, Revit); 3D animation (Maya, Blender); video production (DaVinci Resolve); genomics and bioinformatics analysis.

Relevance:

Engineering, architecture, and CS faculties use workstations for specialist labs. Final year projects involving 3D modelling or ML training need workstation-class hardware.

Personal Computers — Desktops & Laptops

Desktop:

Separate tower and monitor. More upgradeable (RAM, GPU, storage individually replaceable), better cooling, lower cost per performance unit, larger screen options.

Laptop:

Portable, battery-powered all-in-one. Suitable for use in lectures, labs, and fieldwork. Limited upgradeability. Higher cost per performance.

Min spec (CS):

Intel Core i5 (12th gen+) or AMD Ryzen 5 (5000+). 8GB RAM minimum — 16GB recommended. 256GB SSD minimum. Windows 11 or Ubuntu 24.04 LTS.

Refurbished:

Widely available in Zambia and excellent value. Check: battery health (replace if <70%), SSD vs HDD (upgrade to SSD if HDD), legitimate OS licence.

OS options:

Windows 11: best software compatibility. Ubuntu Linux: free, powerful for development, excellent for CS students. macOS: Apple hardware only, excellent Unix environment.

Zambia cost:

New: K3,000–K12,000 for a usable student laptop. Refurbished: K1,500–K4,000 for a reliable 8GB SSD machine.

Mobile Devices — Smartphones & Tablets

6.8B

smartphone users
worldwide (2024)

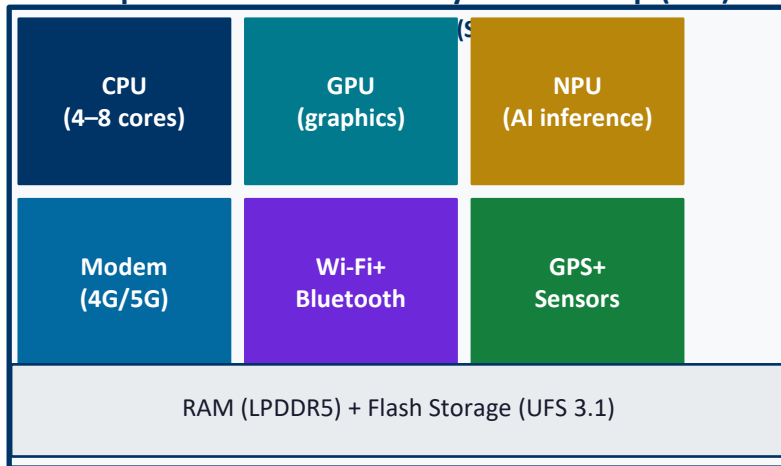
66%

of world population
owns a smartphone

80%+

Zambia's internet
access via mobile

Smartphone Architecture — System-on-Chip (SoC):



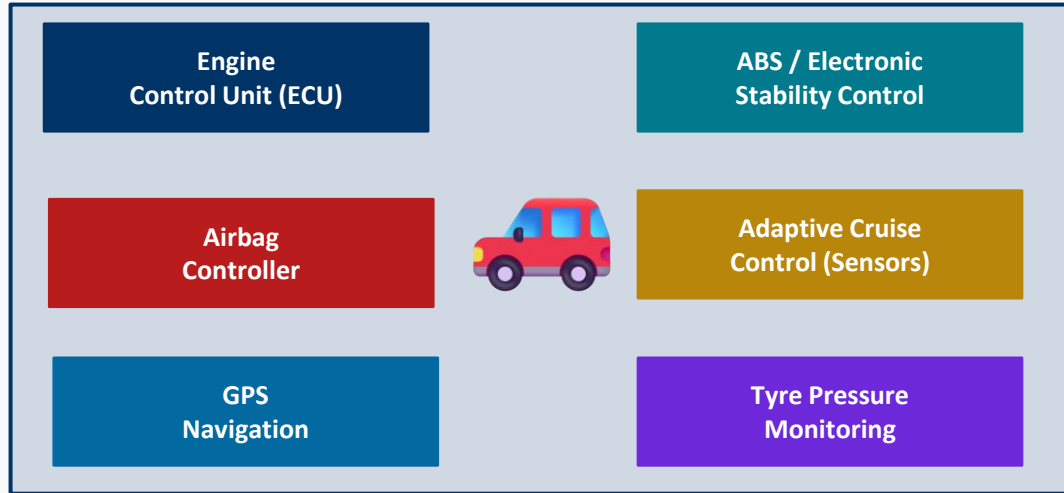
Key Features & Zambian Market

OS:	Android (72% global share): open-source, powers Tecno/Infinix/Samsung in Zambia. iOS (27%): Apple only, high cost, lower Zambian market share.
SoC brands:	Qualcomm Snapdragon (premium), MediaTek Dimensity (mid/budget), Apple A-series/M-series (iOS only), Samsung Exynos.
Connectivity:	4G LTE (most of Zambia), 5G (urban rollout in Lusaka/Kitwe — Zamtel + MTN). Wi-Fi 6, Bluetooth 5.x, NFC (contactless payments), GPS.
Zambia brands:	Dominant: Tecno (budget-mid, African-market-focused), Infinix, itel (entry), Samsung Galaxy A-series (mid-range). iPhone has limited reach due to cost.
Mobile money:	MTN MoMo (*303#) and Airtel Money (*115#) are deeply integrated into Android in Zambia. USSD codes work even on basic feature phones without data.
Tablets:	iPad (iPadOS) and Samsung Galaxy Tab (Android). Used in Zambia for POS terminals, NGO digital health tools (DHIS2 data collection), e-learning.

Embedded Systems — Computers Hidden Inside Products

Embedded System: A computer system with a DEDICATED FUNCTION built into a larger mechanical or electrical device. The user interacts with the product — not the computer. The computer is invisible inside.

FIGURE: Embedded Computers in a Modern Vehicle



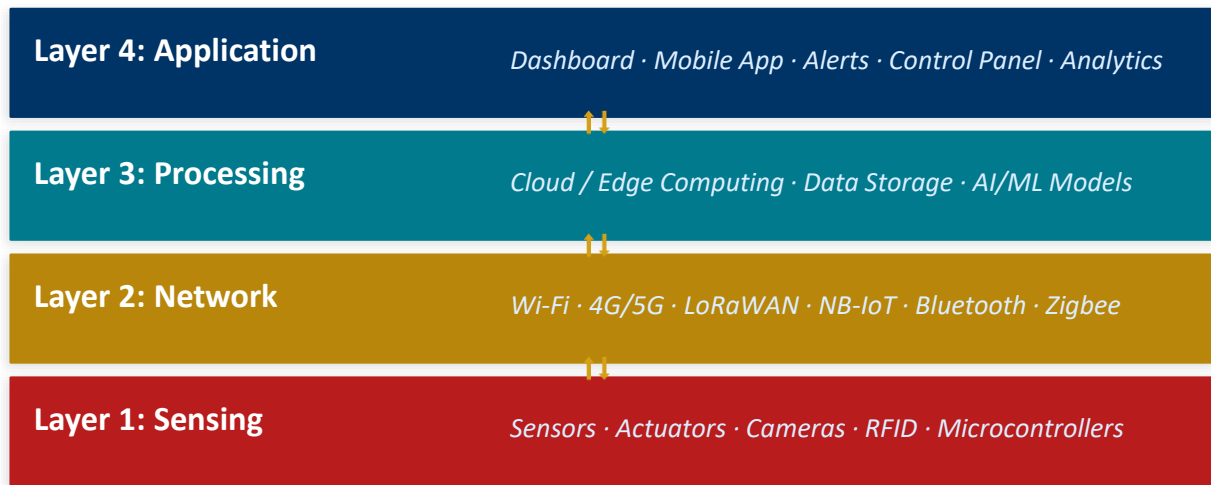
Modern vehicles contain 70–100 embedded computers. An ECU monitors and adjusts engine timing thousands of times per second.

- ▶ Single dedicated function: Designed to do ONE thing reliably — not a general-purpose computer
- ▶ Real-time operation: Must respond to inputs within strict timing constraints
- ▶ Resource constraints: Limited CPU, RAM (often kilobytes), and power compared to PCs
- ▶ High reliability: Cannot fail — pacemaker, aircraft autopilot, vehicle ABS must work every time
- ▶ Invisible to user: You interact with the car/microwave/ATM — the computer is hidden inside

The Internet of Things (IoT) – Computing Enters the Physical World

IoT: A network of physical devices — embedded with sensors, software, and connectivity — that collect and exchange data over the internet WITHOUT human-to-human or human-to-computer interaction. (Campbell et al., 2023)

FIGURE: IoT System Architecture (4 Layers)



Zambia IoT Applications

Smart Agriculture:

Soil moisture & temp sensors on Copperbelt farms → automated irrigation activation

Smart Electricity:

Zesco smart meters → real-time consumption tracking → prepaid top-up via USSD

Mine Safety:

Underground gas sensors + vehicle GPS → evacuation alerts + collision prevention

Health Monitoring:

Maternal health wearables in rural clinics → vitals sent to district health officer

Smart Traffic:

Intersection camera systems → optimised signal timing → reduced congestion

IoT SECURITY WARNING: IoT devices are frequently shipped with default passwords and no encryption. The Mirai Botnet (2016) hijacked 600,000 IoT cameras and routers to execute the largest DDoS attack in history. Zambia's smart meters and mine sensors must be secured — insecure IoT is critical infrastructure risk.

Cloud Computing

"Cloud computing is the delivery of computing services — servers, storage, databases, networking, software, analytics, and intelligence — over the internet to offer faster innovation, flexible resources, and economies of scale." — Microsoft Azure, 2024

Cloud Service Models — The Stack

SaaS — Software as a Service

Provider manages EVERYTHING. User only manages data and accounts.

e.g. Gmail · Microsoft 365 · Zoom · Moodle · Canva · Salesforce

PaaS — Platform as a Service

Provider manages infrastructure + OS + runtime. User manages code and data only.

e.g. Google App Engine · Heroku · Azure App Service · AWS Elastic Beanstalk

IaaS — Infrastructure as a Service

Provider manages physical hardware. User manages OS, middleware, apps, and data.

e.g. AWS EC2 · Google Compute Engine · Azure VMs

Zambia Cloud Examples

SaaS

CBU students use Microsoft 365 Education — Word, Teams, OneDrive. No installation needed. Paid per student by CBU.

PaaS

A CBU student deploys their web app final year project to Heroku. They write code; Heroku manages the servers automatically.

IaaS

A Lusaka startup rents 10 virtual servers on AWS Africa (Cape Town) to run their e-commerce platform — paying only per hour of use.

Computer Classification — Summary Comparison Table

Type	Size	Performance	Cost (USD)	Primary Use	Zambia Example
Supercomputer	Warehouse	ExaFLOPS	\$100M+	Scientific research, AI training	None in ZM; access via cloud HPC
Mainframe	Room	TeraFLOPS	\$1M–\$75M	Banking, government, airlines	Zanaco, Standard Chartered ZM backends
Server	Rack unit	High	\$2K–\$1M+	Web, email, databases, apps	CBU Moodle, ZICTA portal, ZRA system
Workstation	Desk	High	\$2K–\$15K	CAD, video production, science	CBU engineering and CS specialist labs
Desktop PC	Desk	Medium	\$300–\$2K	Office work, education	CBU computer labs, government offices
Laptop	Portable	Medium	\$400–\$3K	Students, mobile professionals	Most common student device at CBU
Tablet	Handheld	Medium-low	\$50–\$1.5K	Reading, POS, NGO data entry	DHIS2 data collection, market POS
Smartphone	Pocket	Medium-low	\$50–\$2K	Communication, apps, banking	Dominant internet device in Zambia
Embedded System	Tiny chip	Low	\$1–\$100	Dedicated appliance control	ATMs, traffic lights, Zesco meters
IoT Device	Micro	Ultra-low	\$1–\$500	Sensing, automation, monitoring	Mine gas sensors, farm soil monitors
Cloud (IaaS)	Virtual	Unlimited	Pay-per-use	Any scalable application	AWS Africa (Cape Town)

Tutorial Discussion Activity – Module 2

Work in pairs or groups of 3 — 12 minutes, then share with class:

1

A hospital in Ndola collects: patient name, NRC, blood type, admission date, hourly temperature readings, nurse's notes, X-ray images, and diagnosis in plain language. (a) Classify each data item as qualitative or quantitative. (b) Classify each as structured, semi-structured, or unstructured. (c) Which items present data quality (CART) challenges and why?

💡 *Blood type is nominal qualitative. Temperature readings are continuous quantitative. What storage system best suits each type?*

DIKW + CART

2

With reference to the Zambia Revenue Authority (ZRA), describe how structured, semi-structured, and unstructured data each play a role in detecting tax fraud.

💡 *When ZRA (Zambia Revenue Authority) wants to detect tax fraud, they must analyse structured financial records (structured), electronic correspondence (semi-structured), and social media posts showing undeclared assets (unstructured). All three types matter.*

DIKW

3

Trace the complete data processing cycle (Input → Process → Output → Storage → Feedback) for Zambia's ZRA e-filing tax system. For EACH stage, identify at least TWO specific technical components and explain what CART quality dimension is most critical at that stage.

💡 *INPUT: What device does a taxpayer use? What data do they enter? FEEDBACK: How does ZRA improve its fraud detection year-on-year?*

Data Processing Cycle

4

A smart agriculture company plans to deploy soil moisture sensors across 200 farms in the Copperbelt. Design the IoT system: (a) Identify the Layer 1 (sensing), Layer 2 (network — justify your choice of connectivity technology for rural Zambia), Layer 3 (processing — cloud vs. edge, justify for intermittent connectivity), and Layer 4 (application) components. (b) Identify TWO specific security risks and preventive measures for each.

💡 *Consider LoRaWAN for rural connectivity. What happens if the cloud connection fails during a drought emergency?*

IoT

Key Takeaways — Module 2

1

Data is raw — information requires context

36.5 is data. '36.5°C — patient temperature, normal range' is information. The DIKW hierarchy requires human judgement to ascend. GIGO: quality in, quality out.

2

Match data type to the right storage and analysis

Qualitative/quantitative and structured/unstructured determine your ENTIRE data engineering approach. 70% of all data is unstructured — and processing it is where the value lies.

3

The data processing cycle is universal

Every computing system — from an ATM to a satellite — follows I→P→O→S→F. The FEEDBACK loop is what makes systems intelligent and adaptive over time.

4

5 generations; we live in the 4th/5th

Vacuum tubes → transistors → ICs → microprocessors → AI/quantum. Moore's Law drove a 12-million-fold transistor increase in 53 years. All your devices are 4th generation hardware.

5

Right computer for the right job

Supercomputers for science, mainframes for banking, servers for networking, PCs for personal work, embedded chips everywhere. Zambia is a mobile-first nation — design for it.

6

IoT and cloud are not the future — they are the present

16+ billion IoT devices. Zambian mines, farms, and utility companies already rely on them. Cloud computing is just servers you rent. Data sovereignty is Zambia's biggest cloud challenge.

Can You Now...? — Learning Outcomes Self-Check

After this module, can you do each of the following without referring to notes?

- ✓ Define Data, Information, Knowledge and Wisdom and give an original Zambian example for each
- ✓ Explain the DIKW hierarchy and describe the transformation required at each level
- ✓ Distinguish between qualitative and quantitative data with 2+ examples each
- ✓ Distinguish structured, semi-structured, and unstructured data — name 2 tools for each
- ✓ Trace the complete data processing cycle for a given Zambian real-world system
- ✓ Apply the CART framework to evaluate information quality with specific examples
- ✓ Name the 5 generations of computers and their key technologies
- ✓ State Moore's Law and calculate approximate transistor doublings over a given period

- ✓ Describe the evolution from mainframes → minicomputers → PCs → smartphones → wearables with Zambian context
- ✓ Classify a given computing device and justify the classification using specific criteria
- ✓ Distinguish a supercomputer from a mainframe: what does each optimise for?
- ✓ Explain what a server is and name 5 types of server with their functions
- ✓ Define embedded system and give 3 specific examples from Zambian daily life
- ✓ Explain IoT: define it, draw its 4-layer architecture, give 2 Zambian applications
- ✓ Distinguish IaaS, PaaS, and SaaS with a real Zambian example of each

Assessment & What's Next

★ This Week's Tasks

Task: Profile a REAL computer (lab PC, home PC, or phone).
Document: CPU (model, cores, clock speed, generation), RAM (size, type), Storage (type, capacity, SSD vs HDD), OS, and full classification. Justify which classification category it belongs to and which generation of hardware.

Module 3 — Coming Next

Number Systems & Data Representation

- Decimal, Binary, Octal, Hexadecimal — conversions
- Binary arithmetic (addition, two's complement)
- Bits, bytes, KB, MB, GB, TB, PB — measurements
- Character encoding: ASCII and Unicode (UTF-8)

Why Number Systems Matter for CS:

- ▶ ALL computer data is stored as binary (0s and 1s)
- ▶ Memory addresses, file sizes, network IPs are all numeric
- ▶ Colour codes (#FF5733), MAC addresses use hex notation
- ▶ Security (encryption), compression rely on binary operations
- ▶ Every CS course from Year 2 assumes binary fluency